

Botlien

Full brief: idea, moat, methodology, and this week's plan

Prepared 2026-08-04 - For Samuel

The Idea

Botlien is real-time collateral risk monitoring for financed and leased commercial service robots. It connects to a robot's own manufacturer telemetry (uptime, location, utilization, fault activity) and turns that into a risk signal for whoever is carrying the financial risk on that robot: a bank financing the purchase, a lender extending credit against a RaaS company's fleet, or the RaaS company itself acting as its own lender.

The core problem: every financed robot already reports its operational health to its manufacturer's cloud. That signal has never been routed to the party actually holding the risk. Right now the first sign of trouble a lender sees is a missed payment, months after the robot itself stopped working.

How it works, mechanically

- Access to a specific financed or leased robot's telemetry is granted as a condition of the loan or lease, the same way GPS tracking and insurance proof are already standard in secured equipment lending.
- Botlien connects to the manufacturer's own API, brand by brand.
- Utilization, uptime, fault data, and status get normalized into one schema across brands, so it stops mattering which manufacturer the robot came from.
- A risk score and early-warning flags get delivered per loan, per lease, or per fleet, continuously.

Who it's for, and the order we go after them

- RaaS operators (primary, first). They own their fleet and carry default/collateral risk directly on every unit they lease out. One buyer in the room, no separate lender to convince first.
- Lenders and credit facilities financing RaaS fleets (primary, first, alongside RaaS operators). Two structures: asset-backed (against one robot's value) and cash-flow-backed (against a RaaS company's whole lease-revenue book, same shape as SaaS venture debt against MRR).
- Robotics manufacturers / OEMs (secondary, later). Deliberately sequenced last, partly because participation is adverse-selected: a manufacturer with reliable robots wants in, one with a rougher record has a real incentive to say no.
- Traditional equipment lenders financing individual robots outright (valid long-run market, not the lead motion right now).

Scope note: Botlien never touches a borrower's financial statements, bank data, or general company financials. Every signal comes exclusively from the robot's own telemetry.

The Moats

Access is contractual, not scraped

Data access comes from a loan or lease covenant written into closing paperwork, not a public API pulled without permission. A competitor can't just build the same dashboard; they'd need the same contractual relationship with the same lender or RaaS company.

Cross-brand normalization is something no single manufacturer would build

A manufacturer has zero incentive to build a tool that also shows a lender how their robots compare to a competitor's failure rate. Working across brands is exactly what makes this valuable, and exactly what one OEM structurally can't replicate.

Lender and RaaS relationships lock in once operational

Once this is written into a credit facility's borrowing-base formula or a lease covenant, ripping it out means renegotiating real paperwork across a whole portfolio. Nobody does that over a marginal price difference.

Two-sided flywheel

Every manufacturer added makes the platform more valuable to lenders and RaaS companies. Every lender or RaaS company added creates more leverage to get the next manufacturer to grant access.

Nobody currently owns this category

Company-level credit bureaus own borrower risk. Fleet telematics companies own hardware-backed asset tracking for vehicles and construction equipment. Nobody owns collateral verification for financed robots specifically, because the category is new enough that no one has built the neutral layer yet.

Technical: What the Real APIs Actually Look Like

This is the part that matters most for you, Sam. As of 2026-08-04 we pulled the real, public documentation for the major manufacturers instead of assuming. Short version: the single hardest technical unknown just got a lot less unknown, at least for one brand.

Bear Robotics - start here

Fully public API reference. No partner application needed just to read the schema. SubscribeRobotStatus streams a consolidated view of robot state in one subscription:

Field	What it tells you
connection.state	Online or offline, right now
battery.charge_percent + state	Charge level, charging/discharging/full
emergency_stop	Software and physical e-stop status
mission	What task is currently running
pose	Current location
error_codes	Active errors, each with a severity level
navigation_state.stuck_state	Stuck status + reason (blocked path, restricted area, unknown)
twist	Linear/angular velocity - is it actually moving

Docs: cloud.api.bearrobotics.ai/v1.3/resources/RobotStatus/ | Getting started: cloud.api.bearrobotics.ai/guides/getting-started/

Pudu Robotics - brand #2, not #1

Real 'Open Platform' program exists. Named categories map well to what we need: Delivery Data, Return Data, Cruise Data, Guiding Data, Cleaning Data, Battery Data, Abnormal Alert Data. But field-level docs are gated behind a 3-step process: become a distributor, apply as a developer, then get platform access. Treat as the second brand once the Bear connector and normalization schema are proven.

Keenon - lowest priority

Minimal public API documentation. Cloud telemetry exists and is user-controlled. API integration is described as 'expanding in overseas markets' (Japan, evaluating Korea), not broadly available yet.

Industry standard worth knowing: VDA 5050

A real, published, manufacturer-independent AMR/AGV fleet communication standard (German VDA/VDMA), MQTT + JSON, now widely adopted. Standardizes position, battery, and driving-status reporting across brands already. MassRobotics AMR Interoperability does something similar. Cross-brand normalization isn't fighting an industry trend, it's riding one.

Humanoid robots - the honest gap

Unitree has no public cloud API at all; SDK-only over local DDS. Agility Robotics has a cloud fleet platform ('Arc') with no public field-level docs found. Figure, Tesla, and UBTECH are earlier still. Humanoid telemetry access is meaningfully less mature than wheeled service robots. This matters directly because IntBot (see below) is a humanoid company - validate the technical bet against Bear first, pursue IntBot on the business side in parallel, not as the technical proof point.

Risk Scoring Methodology

Rules-based first, not machine learning. ML only makes sense once there's enough real historical default data to train against, which won't exist for a long time. A transparent rule set is also what a lender's credit committee can actually audit and trust - a black-box score can't do that.

Signal to flag mapping (using Bear's real schema)

Signal	Source field	Flag rule (v1, unvalidated)
Availability	connection.state	Offline past a set threshold, or rising offline frequency
Utilization	mission history	Active time drops vs. a trailing baseline, per robot type/context
Faults	error_codes + severity	High-severity code, recurring code, or rising error rate
Mobility health	stuck_state, twist	Repeated stuck events, or near-zero velocity during active mission
Battery health	battery over time	Irregular charge cycles or faster-than-baseline degradation

Probability of Default vs. Loss Given Default - kept separate, never blended

- PD-relevant: declining utilization, rising faults, going quiet. A proxy for borrower distress inferred through asset behavior.
- LGD-relevant: current physical condition + depreciated value. Tells you what you'd actually recover if it defaulted.

Collateral value estimate = base depreciated value (age- AND usage-adjusted, since we have real usage data instead of a blind calendar schedule) multiplied by a condition multiplier from the health/fault score. No real robot resale-comp data exists yet, so early estimates stay conservative (manufacturer-stated useful life, straight-line) until real pilot data builds a track record.

Gaps found and fixed across two audit passes

- No empirical validation yet - every threshold is a hypothesis until backtested against real defaults.
- Thresholds need to be calibrated per robot type and deployment context, not one flat number for everything.
- Data-quality layer: track a connector-level heartbeat separately from each robot's own status. One robot going quiet while its neighbors report fine means a real robot problem. Every robot on one connector going quiet at once almost always means a pipe/vendor outage, not a fleet-wide coincidence.
- Systemic vs. idiosyncratic risk: a manufacturer firmware push or cloud outage shouldn't look like hundreds of individual borrowers failing at once.
- Portfolio-level view needed: concentration and correlation by brand, vertical, and geography, not just single-robot flags.
- Time horizon: CECL requires lifetime expected losses, not a point-in-time snapshot. The score eventually needs to project forward over the loan's remaining life, not just report current state.
- Conflict of interest: if a RaaS company pays for monitoring and a lender relies on the same score to underwrite that RaaS company, that's structurally close to the issuer-pays rating-agency model criticized after 2008. Needs a stated answer (separate relationships with disclosed firewalls, or lender-pays for the underwriting-facing product).
- Moral hazard: once money is on the line, borrowers can game a signal (keeping a robot moving just enough to avoid flags). Needs a tamper/integrity check eventually.
- Robot-category-aware rules: a humanoid standing still while working looks identical to a broken

delivery robot under the same rule set. Needs a category layer from day one.

- Bias/fairness testing: real regulatory precedent from usage-based insurance pushes for this before full acceptance. Build it in early, don't retrofit after a regulator asks.

Regulatory Grounding

This is what makes the pitch to a lender a compliance argument, not just an efficiency one.

CECL (Current Expected Credit Losses, FASB ASC 326)

The real US accounting standard governing how lenders must estimate credit losses. Requires a forward-looking, LIFETIME loss estimate, not 'wait until a loss is probable' like the old standard. Explicitly requires considering the nature of collateral and potential future changes in its value. This is the actual regulatory reason a lender needs continuous collateral signal, not optional monitoring.

UCC Article 9

The law governing security interests in equipment as collateral - literally where 'lien' comes from, the root of the company name. Requires lenders to periodically confirm collateral still exists and is in the debtor's possession. Botlien automates an existing legal obligation, doesn't invent a new one.

SR 11-7

Federal Reserve / OCC joint guidance on model risk management, covers third-party vendor models used by banks. A bank remains accountable for independently validating and monitoring any vendor model it uses - can't treat it as a black box. Means the real long-term deliverable is a documented, auditable methodology, not just a working dashboard.

Usage-based insurance / telematics precedent

The closest real analog for novel IoT/behavioral data getting regulatory acceptance into an actuarial risk model. Proof this category of problem has been solved once already, in a closely analogous industry. Regulators there also pushed for bias/fairness audits before full acceptance.

How this becomes an actual interest rate

Botlien should never be the thing that outputs an interest rate directly - that's a scope boundary worth holding firmly. What we build is the input:

Expected Loss = $PD \times LGD \times EAD$ (probability of default, times loss given default, times exposure at default). That feeds the lender's own formula: $APR = \text{cost of funds} + \text{expected loss} + \text{capital charge} + \text{servicing cost} + \text{margin}$. Cost of funds, capital charge, and margin are the lender's own proprietary economics, not ours to know or output. We deliver a continuously updated, well-documented PD estimate (and separately, an LGD-relevant collateral estimate) that plugs into a lender's existing pricing engine.

The IntBot Lead

Antonio met founder/CEO Lei Yang in person and has direct contact info. San Jose humanoid robot company, founded 2024, roughly 12 employees, no disclosed funding found (private data, may be stale). Real commercial traction: a partnership with Certis Group for deployments across Singapore (concierge, wayfinding, multilingual service across transit, hospitality, healthcare, retail), plus a deployment at San Jose Airport.

The one fact that determines the entire pitch

Does IntBot own the robots it deploys with Certis and airports and lease them out (self-financing RaaS, meaning IntBot itself is the natural first customer, no third-party lender needed), or is hardware sold outright to the operator (closer to the original OEM-data-partner model)? Don't pitch anything until this is known.

Discovery call agenda

- How are the Certis and airport deployments structured commercially - owned-and-leased, or sold outright.
- What do they track internally on deployed units today, if anything.
- If self-leasing: is there any early-warning system today before a unit becomes a write-off.
- If sold outright: is there financing or leasing involved on the buyer's side.

NDA before any real technical detail gets shared. Ask for a small, time-bound pilot (a handful of units, 6-12 months) with a first-look or short-exclusivity clause, not a big ask. IntBot being small, early, and unfunded, and needing partnership credibility, makes them realistically reachable in a way a large, resourced OEM wouldn't be right now - but their technical telemetry stack is likely far less mature than Bear's, since humanoid robots generally lag wheeled service robots on public fleet APIs.

This Week

Samuel

- Request real Bear Robotics API credentials and pull a live RobotStatus stream. Fully unblocked, public docs, no gate - the highest-leverage task on the list.
- Structure the connector so Pudu can slot in later without a rewrite, and build the connector-level heartbeat check in from the start, not after.
- Start basic flag logic against whatever data can be pulled: offline duration, error severity, stuck state. Doesn't need to be polished, needs to exist.

Antonio

- Get an NDA ready to send same-day if a call goes well.
- Call Lei Yang at IntBot. Lead with the ownership question before pitching anything. Know both versions of the pitch going in.
- Identify and reach out to 2-3 more RaaS operators or specialty lenders in parallel - don't let IntBot be the only live conversation.
- Reframe talking points around RaaS-first, not lender-first, before those calls - the sequencing changed, the old deck still leads with the original order.

Already handled

- Trademark/domain check on the name Botlien - done.
- Website - in motion separately, doesn't block outreach.

What we still don't know, on purpose

- Whether a manufacturer will grant PRODUCTION access, not just let us read the docs.
- Whether asset/telemetry decline actually predicts default at all - unproven until real pilot data exists. This is the real gating question now, more than API access.
- Realistic scale of small-business robot financing volume today.